



The MANUS-HAND* Dextrous Robotics Upper Limb Prosthesis: Mechanical and Manipulation Aspects

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Abstract. Dextrous artificial hand design and manipulation is an active research topic. A very interesting practical application is the field of upper limb prosthetics. This paper presents the mechanical design and manipulation aspects of the MANUS-HAND project to develop a multifunctional upper limb prosthesis. The kinematics of our design makes use of the so-called underactuated principle and leads to an innovative design that triples the performance of currently existing commercial hand prosthesis. In addition, the thumb design allows its positioning both in flexion and opposition. As a consequence, up to four grasping modes (cylindrical, precision, hook and lateral) are available with just two actuators.

The proposed impedance control approach allows the fingers to behave as virtual springs. Given the difficulty of including the user in the control loop, this approach is based on an autonomous coordination and control of the grasp. As a consequence, the requirements on the Human Machine interface are reduced. At the end of the paper, we briefly describe the clinical trials that were set up for evaluation purposes.

Keywords: dextrous hands, prosthetic hands, EMG control

1. Introduction

The scientific topic of grasping and manipulation with robotics dextrous hands has been active during the last three decades. The idea of having dextrous manipulative, non task specific robotic tools has been behind the interest of a number of research

*The MANUS-HAND prosthesis has been developed under EC contract (DE-4205). Its original name was MANUS. However, it is to be noted that the name MANUS has been also used in the context of a commercial wheelchair-mounted robotic arm. In addition, also the MIT-MANUS therapy robot already exists.

groups world-wide. The areas covered by these research groups range from kinematics and actuation of multifingered robotic hands, to dynamic control of grasping and manipulation including design considerations.

One of the issues most often addressed by the researchers is hand kinematics. The manipulation and grasping capability of multifingered robotics hands has been addressed through the concepts of Mobility and Connectivity of multifingered hands, see Mason and Salisbury (1985). Mobility is defined, in the context of mechanisms composed of serial and parallel links, as the number of independent parameters required to completely specify the position of every body. On the other hand, the connectivity between two bodies in the same mechanism is defined as the number of independent parameters required to specify the relative position of these two bodies. Based on these definitions, the kinematics of the multifingered hand can be defined according to the required task, i.e., when dynamic manipulation is required, a connectivity of $C = 6$ gives the maximum dexterity; when grasping is required, a connectivity $C \leq 0$ ensures that no degree of freedom is left to the grasped object. In the case of hand prostheses, current technology both in the areas of hand design and HMI does not provide realistic means for dynamic manipulation and thus the latter approach is usually attempted, see for instance Kyberd and Chappel (1994).

Actuation has also been a research topic in the field of robotics dextrous hands. The actuator itself was already identified in 1982, in the context of a workshop held at MIT (Hollerbach, 1982), as “the most serious long term impediment to artificial hand design”. In the field of prosthetic hands, the use of underactuated solutions is widespread. In underactuated hands, the number of motors is lower than the number of active joints, so that some kind of joint motion coupling should be provided. For light solutions, generally two approaches are used, namely, bar linkages and tendon or cable transmission. Vinet *et al.* (1995), presented a particular solution based on bar linkages to link the motion of the first and second phalanges of a prosthetic hand. The second solution has been applied in a number of hand designs, see for example, Liu *et al.* (1998), Butterfass *et al.* (1998) and others.

Underactuated hands have the advantage of system simplicity, i.e., the number of required actuators is reduced while preserving the number of active joints. This, however, leads to fixed transmission ratios be-

tween joints and a reduction of adaptation capability to object geometry.

Whenever a rigid coupling between phalanges is used, no adaptation to object geometry is achievable. In order to cope with this lack of adaptability, compliant transmissions between joints have been applied. This second approach was already introduced by Hirose and Umetani (1977) in a design with tendons that can adapt to object geometry. Likewise, in the context of the Southampton hand, the underactuated principle is applied by using a whiffle-tree mechanism between the index and middle fingers. This approach allowed for independent compliant adaptation of each finger to object geometry (Kyberd *et al.*, 1998).

A third topic of research within the area of grasping with dextrous robotics hands is hand control. Hand control typically covers grasp pre-shaping and grasp synthesis operations. The first problem, grasp pre-shaping appears after the difficulty of generating contact location points from a simple description of the object to be grasped, the hand and the task. Under these circumstances, grasp pre-shaping is used as a grasp pre-planner and a large amount of work has been devoted to this topic, see Shimoga (1996) for a review. To the authors’ knowledge, in the framework of hand prostheses, grasp pre-shaping is only used by the Southampton Adaptive Manipulation Scheme, SAMS. In this approach, contact sensors distributed in the hand are used for a first pre-shaping operation between the two available grasping modes, i.e., precision and power, see Kyberd and Pons (2003).

This background has been mainly developed in the framework of robotics dextrous hands. The aim of obtaining general purpose end-effectors has been behind all this effort. Nevertheless, as Pfeiffer (1996) pointed out, one of the most interesting and practical background for the development of dextrous hands are prosthetic applications. Current prostheses have a reduced set of active joints (generally 1 or 2, see Otto Bock System Electric Hand, 2001). If dexterity is to be increased, a larger number of active joints are required. In addition, this has to be compatible with reduced size and weight for aesthetic solutions.

In this paper we introduce the approach of the MANUS-HAND project towards the design and manipulation with a multifingered dextrous hand for upper limb prosthetics. In a first section we will discuss the overall approach from specification to system clinical validation. In a second section we will detail the mechanical approach to the design of the prosthesis

while in the third section we will focus on the manipulation approach including EMG management, control architecture and strategy. We will end up with a discussion on results.

2. The Concept of a Multifunctional Upper Limb Prosthesis

After the study of the taxonomy of human hand, see Pons et al. (1999) and Reynaerts (1995) for a review, the manipulative movements of the hand can be classified into extrinsic and intrinsic movements. Extrinsic movements are those of the grasped object by displacement of the hand as a whole, while intrinsic movements define the motion of the prehended object within the hand. The former is related to object grasping while the latter is related to dynamic manipulation of the object.

Once the task is specified, i.e., either grasping or dynamic manipulation, the requirements for the kinematic structure, the number of active joints and for the control strategy follow. When dynamic manipulation is considered there is a consensus that at least 3 or 4 independently actuated fingers are required. However, this requirement is much lower when only grasping is considered as the target task. In fact, if extrinsic movements of the hand are considered, approximately 90% of the human hand capability can be accomplished with four basic grasping modes, i.e., cylindrical, hook, tip and lateral.

The MANUS-HAND project was set up in 1998 to develop a multifunctional robotic hand prosthesis with the aim of obtaining at least these four grasping modes. In our approach, we were conscious of the importance of involving potential users in the specification of the hand. Therefore, the MANUS-HAND approach followed a methodology in which the system was specified according to potential users, i.e., upper limb amputees with different levels of amputation, and rehabilitation professionals.

The overall approach comprised a three-fold modular path:

1. The system should allow different types of grasping modes according to the user needs.
2. The system should be applicable regardless of the residual EMG activity of the muscles of the amputee.
3. The system should allow a stepwise extension to the shoulder.

2.1. System Specification

As stated above, the system specification was an interactive process taking into account the users and rehabilitation professionals' opinion. In order to do so, the clinical partners in the MANUS-HAND consortium prepared two sets of questionnaires that were submitted to 200 amputees and 150 rehabilitation professionals respectively; roughly 60% of the questionnaires were answered and led to the specification partially shown in Table I. Table I represents average figures for the different specification categories. The last column in this table, shows the comments on the technical specifications after a feasibility analysis. The term "Accepted" refers to the situation in which minimum and desired specifications were considered technically feasible within the scope of the project.

The questionnaires tried to identify the relevant functionality of the hand prosthesis with regard to real use activities. The results show the relative importance of the different aspects involved in the manipulation with hand prostheses as perceived by both amputees and therapists. They can be summarised as follows:

- The main concerns of the amputees are aesthetics (62%), discomfort (58%), excessive weight (58%), lack of functional capabilities (50%), problems with technical assistance and maintenance (31%), noise (27%) and problems with the skin of the stump (irritation, sweating . . .) (15%).
- On the other hand, the rehabilitation professionals focused on the functional requirements. In their opinion, the prosthesis should provide the five main types of grasping, namely, tip grasp (55%), hook grasp (50%), precision grasp (45%), cylindrical grasp (35%) and lateral grasp (30%). When we asked about the importance of the sensorial information, most of therapists answered positively (88%), and consider the best biofeedback would be tactile (59%). Based on their opinion we identified the technical requirements of the hand prosthesis. The technical requirements were stated in terms of required grasping modes, required opening span and maximum force for long term grasps for every grasping mode, weight, size, system autonomy . . . The definition of grasping modes is compliant with the taxonomy of human hands due to Schlesinger (1919). Figure 1 schematically depicts the grasping steps according to the above mentioned grasping modes.

Table 1. Technical specification of MANUS-HAND prosthesis prototype

Feature	Minimal	Desired	Tech. Evaluation
Functional specifications			
Opening span	100 mm adults 60 mm children	100 mm adults 60 mm children	100 mm adults (only one universal prosthesis)
Opening velocity	15 to 55 mm/sec	110 mm/sec	Accepted
No. of active fingers	3	4	Accepted
Prono-supination <i>range</i>	170 degree, ± 7 degree accuracy	170 degree, ± 7 degree accuracy	Accepted
Prono-supination <i>velocity</i>	20 to 60 degree/sec	10 to 90 degree/sec	Accepted
Prono-supination <i>torque</i>	2 N * m	3 N * m	Accepted
Modes of grasping	Hook, cylinder, fine grip	hook, cylinder, pitch, fine grip, lateral	Hook, cylinder, fine grip & lateral
Power and characteristics of grasping	15 N One grip mode For children the maximal grip force is between 8 to 50 N	60 N Differential grasping between various fingers. The thumb must be able to carry out opposition. The thumb may act independently by a metacarpophalangeal joint with range of 0 to 70 degree, and interfalangeal joint range of 0 to 40 degrees	Accepted, differential grasping decision left for a more thorough study
Flexion-extension		90 degree, 60 degrees/second, 3 N * m torque	No active flexion-extension of wrist
Lifting	100 N	140 N	
Locking position: Hand wrist flexion/extension prono-supination power/gripping for each position	One "stop and freeze" position; Default two intermediate positions	One "stop and freeze" position; Default two intermediate positions and one programmable default position	Accepted
Cosmetics specifications			
Weight of the prosthesis, including cover and battery for adults (children)	850 g (200–300 g)	400 g (200–400 g)	Maximum 800 g + belt battery up to 500 g
Cosmetic gloves disturbance to pronation/supination movement	Limited disturbance	No interference	Accepted
Noise during activation	No noise	No noise	Accepted
Performance specifications			
Build-in battery tester			Alarm if charge is lower than allowed
Battery position	On the prosthesis	On a trouser belt	Both options combined
Charging time	12 hours	6 hours	Option to recharge through the car special socket
Battery	6 V, 1000 mA/H, 200 cycles, 200 gr weight	6 V, 1300 mA/H, 400 cycles, 165 gr weight	To be studied

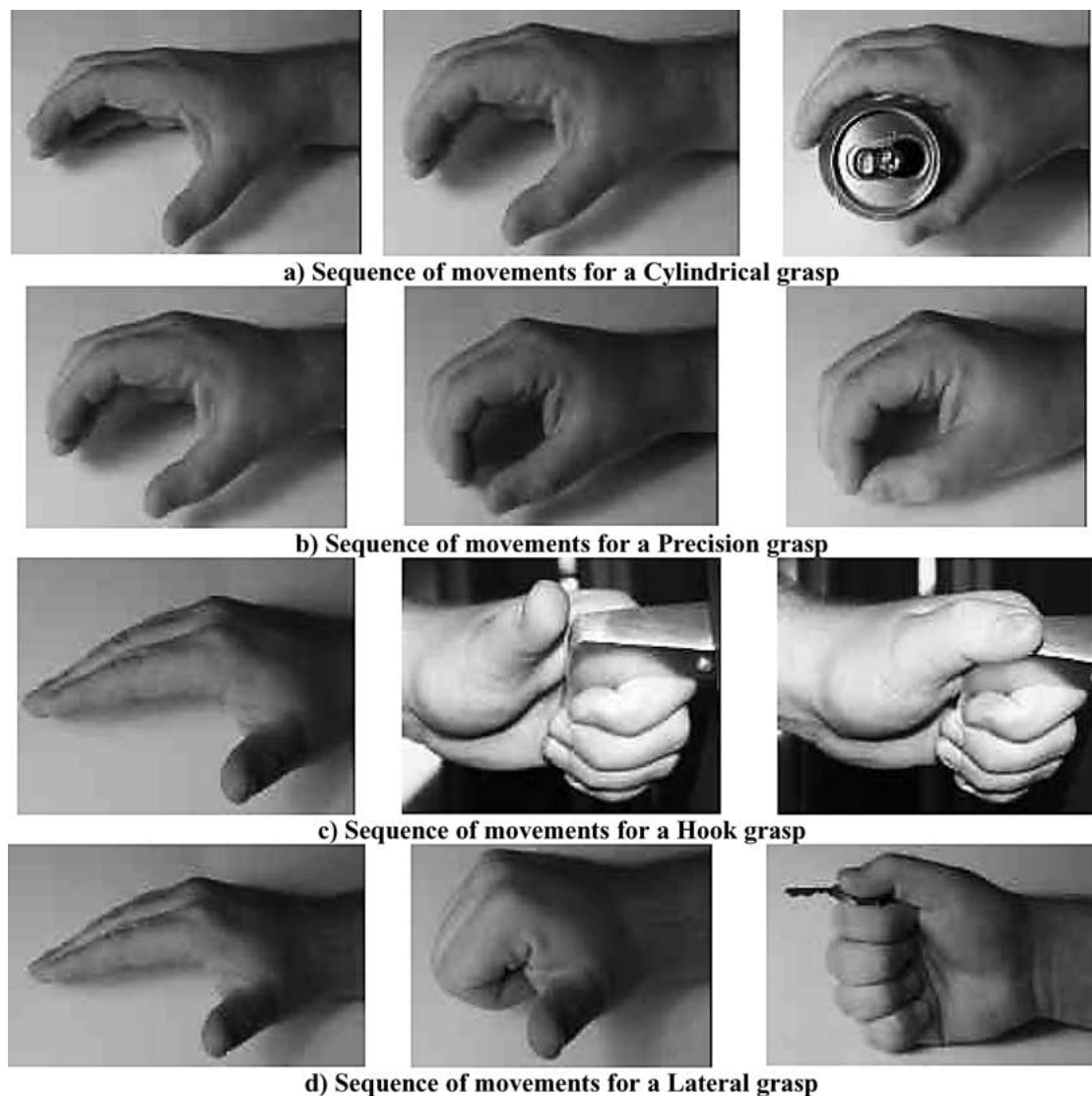


Figure 1. Grasping modes allowed by the MANUS-HAND prosthesis and corresponding kinematic sequences: (a) Cylindrical grasp, (b) precision grasp, (c) hook grasp, and (d) lateral grasp.

2.2. Prosthesis Prototype: Design & Manufacturing

The above described specification lead us to a conceptual design stage in which technological solutions for the different prosthesis subsystems were evaluated. The different possible approaches were technically assessed and, eventually, a final design for every prosthesis subsystem was selected. In the next sections, the mechanical and manipulation approach will be developed in more detail.

After the design stage, the prototype was manufactured, assembled and tested in laboratory conditions.

Following this process, a clinical evaluation protocol was defined for functional validation, this is described in the next paragraph.

2.3. Clinical Evaluation

The clinical evaluation in real use conditions is the key test for validation of any technical aid for disabled and, in particular, for an upper limb prosthesis. The MANUS-HAND consortium set up clinical trials in order to validate the hand prosthesis. The clinical trials were conducted at rehabilitation centers in Spain and

Israel, The CRMF-IMSERO and The Flieman Hospital respectively.

The patients involved in the clinical trials were selected from the users answering the specification questionnaire at the beginning of the project. All the subjects were informed of the clinical evaluation protocol and written consent was obtained from them. At the end of this paper, the results are discussed in detail.

3. Mechanical Design

According to the technical specification conducted from the opinion of users and therapists, several mechanical solutions were analyzed. They will be described in detail in this section. The mechanical concept contains several innovations with respect to state of the art prostheses:

- the number of axes is more than tripled as compared to commercial prostheses, see Otto Bock System Electric Hand (2001). However, the use of additional active joints has been considered in the context of research hands by a number of authors.
- a crossed-tendon mechanism is used in the fingers instead of the more traditional bar mechanism in the field of robotics. Again, commercial prostheses only provide fixed shape fingers. Research prototypes are found in which each inter-phalangeal joint is motor driven, see for instance the DLR hand Liu et al. (1998) and Buterfass et al. (1998). Other approaches, see Kyberd and Chappel (1994) use four-bar mechanisms to provide a natural flexion pattern.

- the thumb movements are coupled by means of a Geneva-wheel based mechanism, which makes it possible to use one actuator for thumb movements in two planes. To the authors' knowledge, this is an innovative aspect of the MANUS-HAND.
- commercial Ultrasonic motors have been used to drive the wrist pronation-supination. Moreover, we have developed our own prototypes of ultrasonic motors with improved performance. They are being build as an alternative to this commercial solution.
- the fourth and fifth fingers are provided with a martensitic structure. These fingers can be manually shaped for long-term grasps. The use of the martensitic structure allows a much higher number of bending cycles as compared to commonly used materials, thus improving the overall reliability of the hand.

3.1. The Kinematic Concept

MANUS-HAND proposes a prosthesis having ten joints of which three are independently driven. The other joints are coupled to the driven ones by different types of mechanisms, according to the underactuated principle. Three independent mechanisms can be found in the MANUS-HAND prototype:

1. the finger mechanism,
2. the thumb mechanism, and
3. the wrist mechanism.

Figure 2 schematically shows the total kinematic concept in which independent joints and coupled ones are identified. In addition to the active joints, bendable joints have been included in the fourth and fifth fingers.

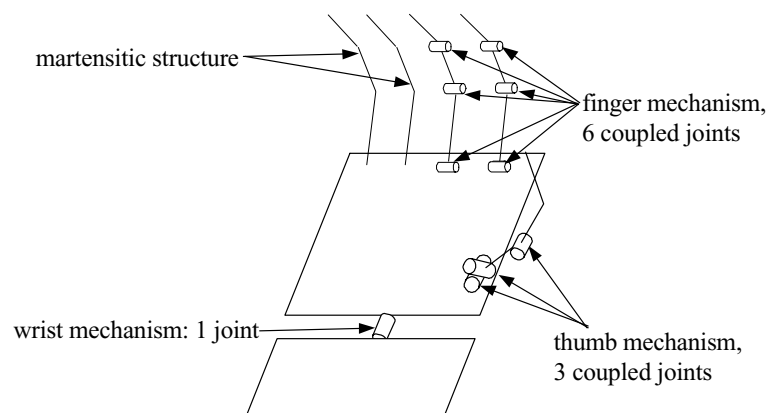


Figure 2. The global underactuated kinematic concept: Up to ten active joints are driven by just three motors.

The use of a martensitic material is the basis of these two fingers. Each finger includes two strips (4 mm wide and 1 mm in thickness) to implement bendable joints. This way, the shape of the fingers can be manually positioned for long term grasps or aesthetics. In the following paragraphs, the finger and thumb mechanisms are described in detail.

3.1.1. The Finger Mechanism. Commercial prostheses use fixed shape fingers. The fingers only rotate about the proximal joint and have been shaped to resemble a human finger as much as possible. This has two disadvantages: from a cosmetic point of view, it is impossible to make the finger look like a human finger for all angles of the proximal joints; secondly, from a functional point of view, a finger with a fixed shape limits the grasping functionality.

The MANUS-HAND concept proposes fingers with three coupled joints. By Coupling the joints, a relationship is imposed between the angles of the phalanges, as a consequence, a certain flexion pattern is obtained. In the MANUS-HAND prosthesis a transmission ratio of 1:1.2 is implemented between consecutive joints. If the resulting kinematics is compared to that of fixed shape fingers, this concept improves both the aesthetics and the functionality of the prosthesis. Since this concept allows an adaptation to the object shape, the resulting required grasping force is lower as compared to fixed shape approaches.

The index and middle fingers are actuated by means of one single motor, so that the proximal joints of the two fingers are linked. In addition, both fingers have three joints. In robot hands, inter-phalanx coupling is traditionally done by means of bar mechanisms (Fig. 3a). When bar mechanisms are used, singular joint positions are found for which only a small lever arm left. Therefore, the force needed to generate the required torque is very large.

This problem can be solved by using a crossed-tendon mechanism, as shown in Fig. 3b. The crossed-tendon mechanism makes the lever arm independent of the finger angle and thus the required force decrease. In the following paragraph, the working principle is shortly described.

Pulley 1, to which the proximal phalanx is attached, is driven by the fingers motor, while pulley 2 is fixed to the palm structure and keeps motionless during grasping.

The distal phalanx of this joint is attached to pulley 2 and tendon 2 is used to connect pulleys 2 and 3. By

means of the crossed pattern implemented with tendon 2, the pulley 3 and the distal phalanx attached to it are forced to rotate with respect to the proximal phalanx. The same scheme is repeated between the medial phalanx and the finger tip.

A manual tensioning mechanism is implemented for each joint. To the authors' knowledge there is no commercial prosthesis in which three joints are linked by means of crossed-tendons, thus this is considered to be an innovative design, even when as above reported, research prototypes can be found that implement both driven joints and other coupling approaches.

3.1.2. The Thumb Mechanism. In order to minimize the number of actuators, and thus the weight and required space, we adopted the solution of having one single actuator for the thumb motion. To this purpose, the movements are transformed into a cycle by means of an intermittent mechanism, the so-called Geneva wheel. In this cycle two movement planes are defined:

1. The first plane corresponds to cylindrical and tip grasps, i.e., thumb is flexed in an opposition pattern.
2. The second plane, implements hook and lateral grasps, i.e., thumb is flexed in a non-opposition pattern.

The key position in this cycle is the neutral position, at which the movement can change from one plane to the other. Figure 4a shows a scheme of the cycle, while in Fig. 4b we show the hand at different representative positions during the cycle. The cycle starts at the lateral grip, with the motor rotating clockwise. While the grip is opening, the thumb passes the hook grip position. At any point during this motion, the direction of rotation of the motor can reverse, so the grip closes again. If the motor continues rotating in the same direction, finally the thumb reaches the neutral position. Further rotation causes a change of the movement plane and the thumb reaches a cylindrical grip. While the motor keeps rotating in the same direction, the cylindrical grip closes and depending on the position of the fingers a precision or cylindrical grasp can be reached.

At any point, the direction of rotation can be reversed by the user and the grip opens again. As mentioned before, the implementation of this thumb cycle is done by means of a Geneva-based mechanism. Figure 5a shows the basis of this mechanism. The Geneva wheels are implemented as gears. While the teeth of both gears are engaged there exists a 1:1 coupling between the two wheels. At a certain point, the gears are no longer in

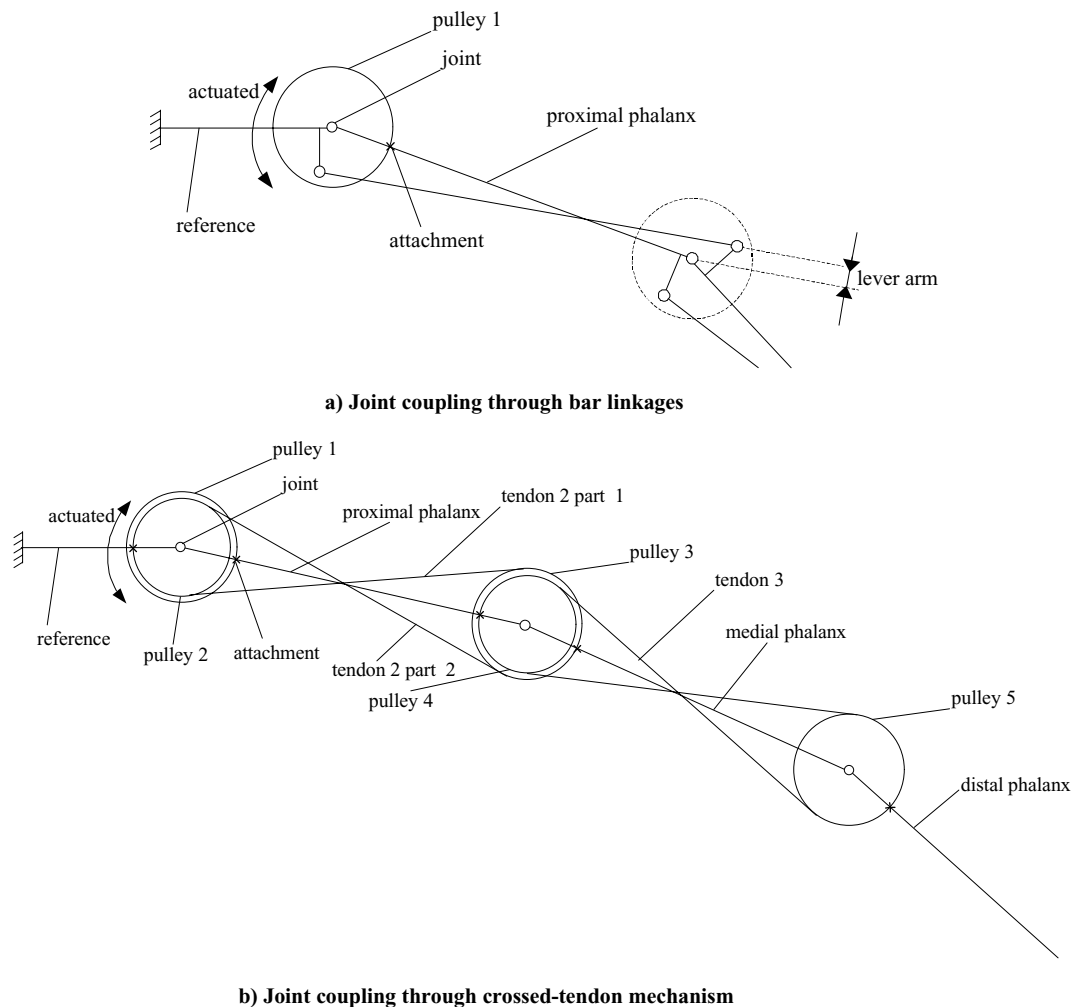


Figure 3. Concepts for underactuated finger mechanisms: (a) coupling through bar linkages and (b) constant torque approach based on a crossed-tendon mechanism.

contact, but the following wheel is locked in its position by means of a form-closed lock. The driving wheel can continue rotating and drive another axis, while the axis connected with the following wheel of the Geneva mechanism is locked. Figure 5b shows the principle of the designed mechanism with two perpendicular axes.

Two drive paths exist in the Geneva mechanism:

1. The first one drives the first axis (cylindrical plane). It starts at the actuated axis and the motion is transmitted by means of gear set 4-3 (transmission ratio 1) and gear set 2-1, see Fig. 5b.
2. The second drive path also starts at the actuated axis. The motion is transmitted by means of the Geneva

mechanism. This means that the second axis (lateral plane) can be coupled to the driven axis (transmission ratio 1) or can be locked. In the first case, the opposition axis and gears 2 and 3 rotate at the same speed. Because the first axis (cylindrical plane) is mounted on the second axis, there is no relative motion between gears 1 and 2 and the thumb does not rotate about the first axis. Thus, the thumb moves in the lateral plane.

In the second case, the Geneva mechanism is locked and the second axis does not rotate. Because of this also the housing stands still and there is a relative motion between gears 1 and 2. This means the first axis is actuated and the thumb moves in the cylindrical plane.

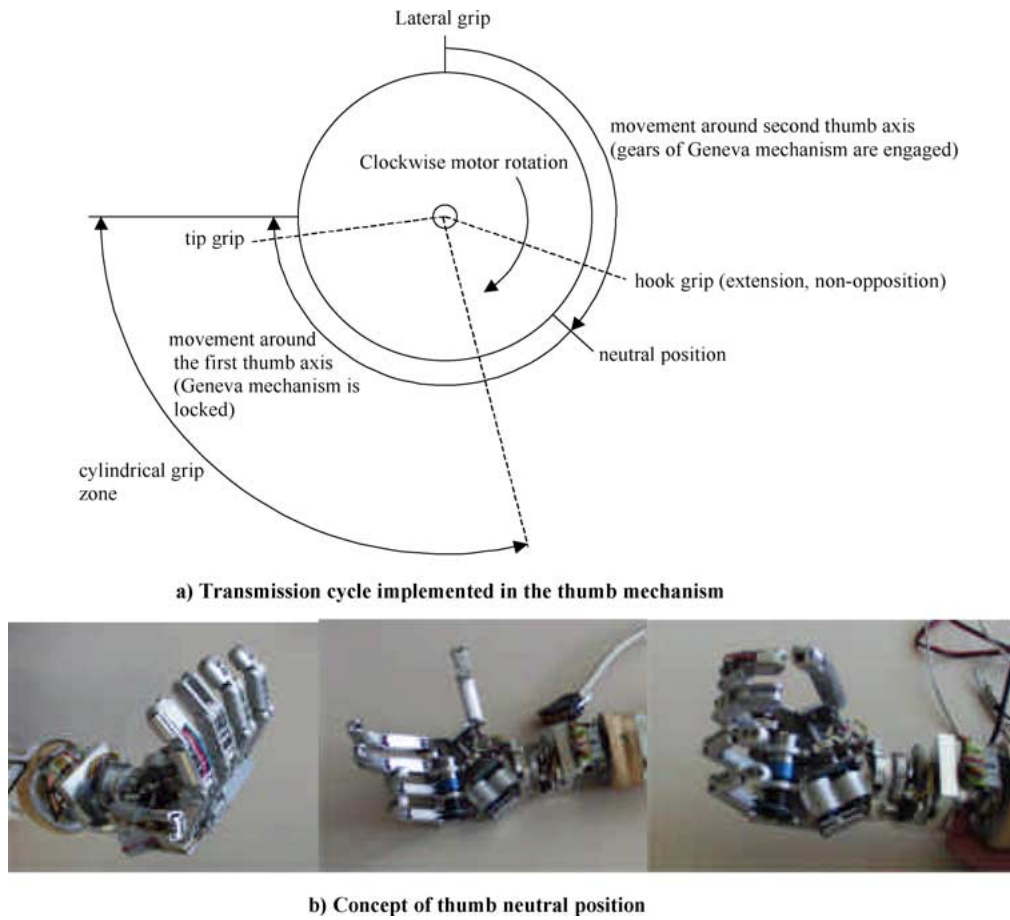


Figure 4. Description of the thumb concept: (a) cycle implemented via a Geneva based intermittent mechanism and (b) corresponding thumb position during lateral grasp, neutral position and precision grasp.

The thumb concept enlarges dramatically the number of possible grips and is not used in commercial available prostheses.

3.2. Actuator System

When considering prosthetic applications, the actuator system becomes very important. The multifunctionality of the prosthesis strongly depends on actuators' performance. According to the technical specification, the actuator will determine to a large extent the autonomy of the entire system, the available grasping force, the weight and size, and aesthetic issues like noise. In order to evaluate the quality of the actuation types within the MANUS-HAND project, 11 properties were selected. Table 2 lists the relative importance of each property with regard to prosthetics. All properties are briefly explained.

- *Efficiency.* The performance of the actuation type is the mechanical output power, divided by the electrical, pneumatic or hydraulic input power. Only the performance of the actuator is brought into account (the transmission and energy converter are brought into account elsewhere). The performance of the actuator is important because of the limited energy storage capacity and the thermal heating.
- *Bandwidth.* Under this category, both bandwidth and response time are considered, even though both properties are closely related. On the one hand, a minimum response time and speed is required to ensure the hand prosthesis closing within less than 1 second. On the other hand, in case an anti-slippage reflex would be included, bandwidth becomes relevant.
- *Energy storage capacity.* Both weight and volume of the energy unit (and converter) are considered. Also the performance of the converter is considered. This

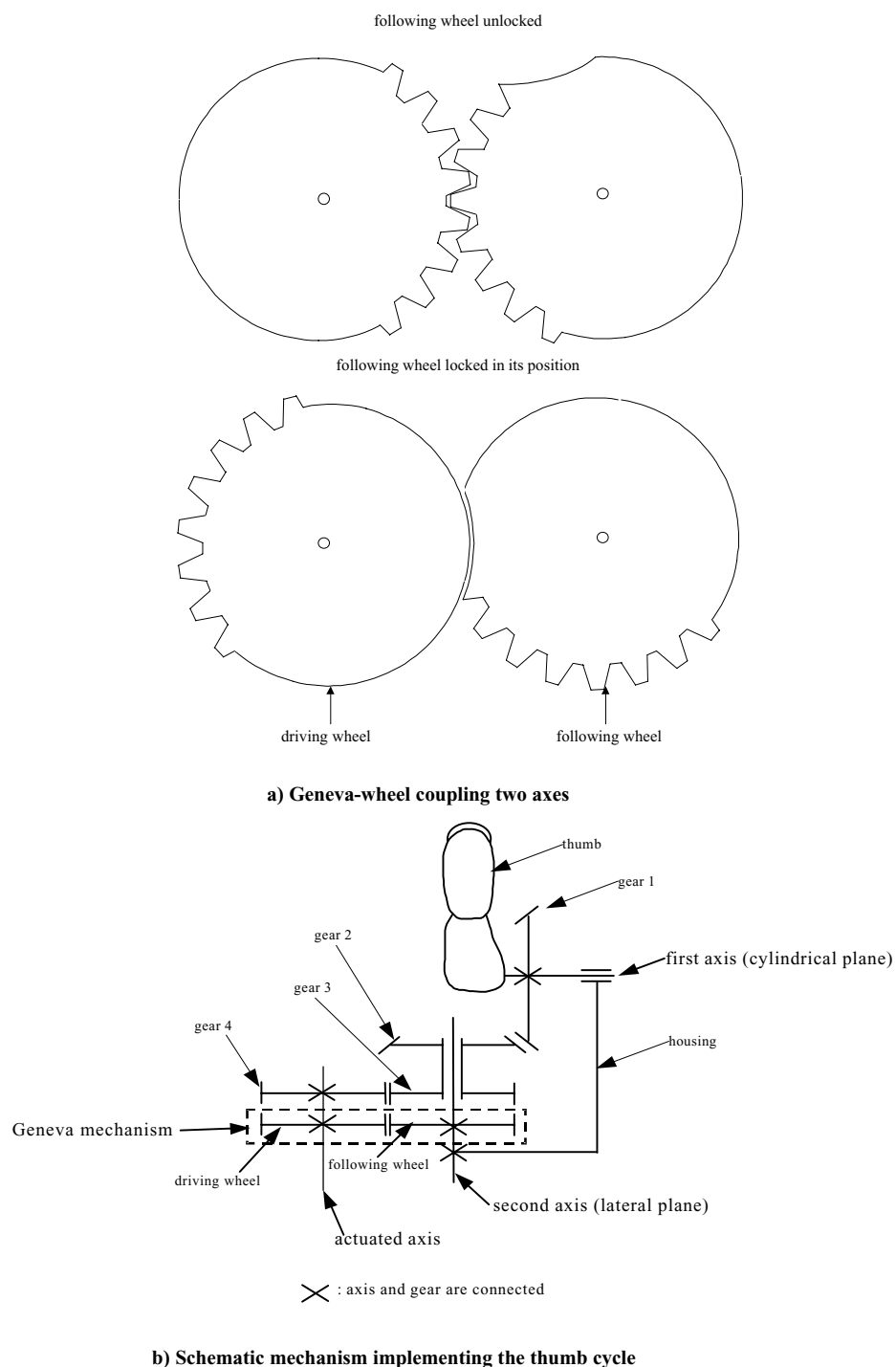


Table 2. Importance of actuation properties.

Properties	Importance
Performance	Very important
Energy storage capacity	Very important
Specific power	Very important
Required transmission	Very important
Self braking	Very important
Noise	Very important
Safety	Very important
Robustness	Important
Volumetric power	Important
Bandwidth	Less important
Controllability	Less important

property is very important because of the limited volume and weight specifications.

- **Robustness.** Robustness is the generic term for all aspects of lifetime, maintenance . . . of the hand prosthesis. Although repairing some parts of the hand prosthesis from time to time is not excluded, it should be avoided as much as possible. Aspects related to resistance to dirty environments were also considered under this topic, even though this is an aspect more closely related to transmissions than to the actuator itself. With regard to this last issue, the use of cosmetic gloves was considered to provide the same degree of protection to any kind of actuator technology.
- **Specific power.** The specific power of an actuation type is the ratio of the mean power to the mean weight of the actuators. However, since speed was included indirectly in the evaluation of bandwidth, here we looked especially to specific force.
- **Volumetric power.** The volumetric power of an actuation type is the ratio of the mean power to the mean volume of the actuators. A similar remark to that of the previous property is applicable to volumetric power.
- **Required transmission.** As a result of energy losses (friction, deformation of gear teeth . . .), the transmission ratio should be kept as small as possible to preserve a good performance. Also weight and volume of the transmission depend on the required transmission ratio.
- **Selfbraking.** Self-braking enables the actuator to deliver a stall torque (force) without energy consumption, (commonly referred to as backdrivability). This self-braking property depends only on the actua-

tion type and not on the transmission type. Blocking the actuators after a stable grasp is made at the expense of lower grasp robustness, in particular in those matters related to coping with grasp disturbances. This is closely related to the high impedance commonly found at the hand-object interface. However, in MANUS-HAND a relative low object-hand impedance was experienced, mainly due to the low stiffness of force sensors, the relative compliant crossed tendon transmission and the rubber glove. In addition, the continuous operation under impedance control was not a realistic alternative due to corresponding high energy consumption it would represent.

- **Noise.** Noise produced by actuators should be reduced to a minimum for comfort purposes. According to our specification process, noise is seen as a major problem by roughly 25% of the users. In some cases, noise is even reported to have the beneficial effect of providing an indirect means of force biofeedback.
- **Controllability.** Controllability of the actuation type corresponds to the ease in which the speed and position of the actuator can be accurately achieved. It contains both the algorithms and the hardware.
- **Safety.** Because of continuous human contact with the hand prosthesis, safety is of paramount importance. The choice of actuator type determines the requirements in terms of driving voltages, i.e., Ultrasonic motors require voltages in the order to 100 V RMS. The efficiency of the actuator is directly imposing the amount of heat being generated, which could also result in skin problems. The amount of force the hand can exert on soft tissues is also a safety factor. However, while in commercial prostheses this can be very high, the use in MANUS-HAND and other hand prototypes of driven inter-phalanx joints reduces the requirements on finger tip force. Therefore, this aspect was not included in the analysis of actuators in terms of safety.

The actuation types were compared by using the Dominic method [16]. The Dominic method starts by combining Table 2 and Table 3 into Table 4. The first column of Table 4 contains the quality code (++: excellent, +: good, 0: sufficient, -: insufficient, --: poor). The first row of Table 4 contains the importance of each property. The letters in the columns 2 to 4 correspond to the actuation types of Table 3. The quotation of every actuator regarding the various different properties was

Table 3. Quality of actuation types.

Motor	Performance %	Bandwidth (Hz)	Energy storage capacity	Robustness	Specific power (W/kg)	Volumetric power (W/m ²)	Required transmission	Self braking	Noise	Control	Safety
A SMA	--	--	+	-	0	0	--	0	++	-	-
B Pneumatic cylinder	0	+	-	+	+	+	0	+	-	-	0
C Hydraulic cylinder	+	0	--	0	+	+	++	++	0	0	0
D Pneumatic motor	0	+	-	+	0	0	-	+	-	-	0
E Hydraulic motor	+	0	--	0	+	++	+	++	0	0	0
F Ultrasonic motor	-	++	0	-	0	0	+	++	+	-	-
G DC motor	+	+	++	+	-	-	0	0	0	++	++
H Brushless DC motor	+	+	++	++	0	0	0	0	0	+	++
I Stepper motor	0	+	++	++	-	-	+	+	0	++	++

Table 4. Trade-off between different actuation principles.

	Very important	Important	Less important
Excellent	ACCEFGGHHII	EHII	FGI
Good	ABBCCDEEEFFFGHII	BBCDGI	BDGHHI
Sufficient	AABBCCDDDEEFFGGGHHI	ACDEFH	CCEE
Insufficient	ABDDDDFGI	AFGI	ABDF
Poor	AACE		A

carried out according to technical sheets when available. For some actuator types, i.e., SMA (A), no commercial data was available and thus laboratory tests together with simulations were carried out to estimate the actuator properties.

Actuation type H (Brushless DC motor) is the only actuation type that always remains above the threshold in Table 4 (dashed line). Therefore, this actuation type H is preferred. Because the self braking capabilities of ultrasonic motors (F), they were also considered an interesting alternative to Brushless DC motors. Brushless DC motors (Minimotor 1727 U006 S-123) were used to drive the active fingers.

Even though ultrasonic motors were ranked in the second position, some application related reasons made us to use also this actuator technology for the wrist rotation. In particular, the strict requirements on actuator size (mainly thickness) when trying to apply the prosthesis to cases of amputation with wrist disarticulation was a determinant factor.

A great deal of work was spent during the development of the MANUS-HAND prosthesis in improving the performance and safety of ultrasonic motors. In

particular, prototypes of double stator ultrasonic motors were attempted, but results were not satisfactory in terms of efficiency. Eventually, we decided to include commercial ultrasonic motors and Shinsei USR30 were selected for the actuation of the wrist pronosupination. Future versions of MANUS-HAND will include our own US motors.

Figure 6 shows the MANUS-HAND prototype and in particular the placement of actuators, transmissions and the Geneva mechanism. Two-stage transmissions were used both for the finger and the thumb mechanisms. The first transmission stage is the standard commercial choice corresponding to the selected Minimotor drives. The second transmission stage is implemented through the Geneva based mechanism in the case of the thumb, and though the crossed tendon mechanism in the case of the index and middle fingers.

In the case of the wrist mechanism, a very low transmission ratio was used (1:10). It was completely designed in order to provide a hollow structure to allow wiring. The ultrasonic motor was connected to the transmission mechanism by a flexible joint so that slight misalignments were allowed.

4. Hand Manipulation

Force controlled grasps require exact computation of equilibrium conditions. Such equilibrium computation relies on feedback information about position of the contact points as well as on a precise estimation of the grasping forces. On the one hand, measuring contact point force for a whole hand grasping operation requires a large amount of force and contact sensors,

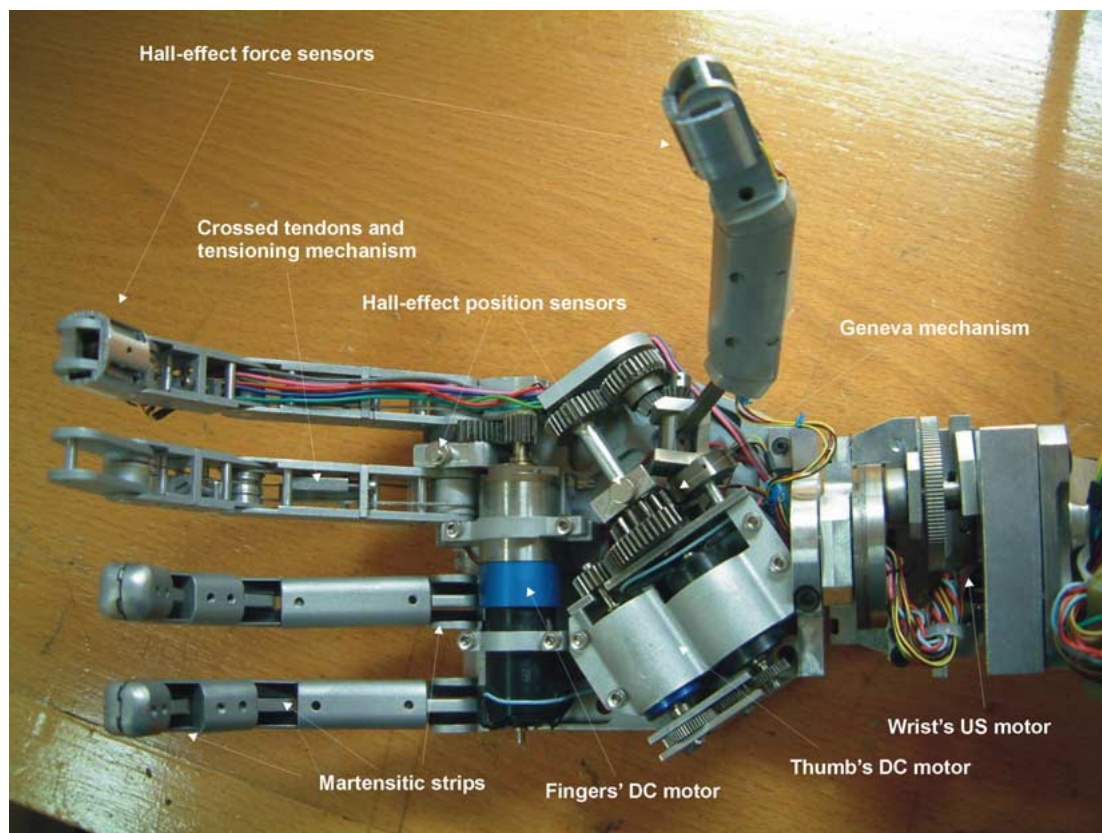


Figure 6. Detailed view of the MANUS-HAND prosthesis: position of actuators, transmissions and tensioning mechanisms, sensors and martensitic fourth and fifth fingers.

making the whole system complex. On the other hand, no previous information is available on object geometry and on location of contact points. In addition, at this stage, no Human-Machine Interface, HMI, would support the bi-directional communication of grasp information required for force controlled grasps.

As a consequence, our control loop does not include the user. The user's EMG signals are used off-line to command grasp modes, and the hand controller implements stable grasps based on a stiffness control strategy.

In order to keep system complexity low, we have studied the possibility of implementing stable grasps based on actively programmable finger compliance (Van Brussel et al., 1989). In this approach, the user would command the desired type of grasp and "pre-shape" the hand; and the prosthesis would autonomously perform the grasp without additional information. This approach would require a HMI supporting a high number of commands to drive a multifunctional hand, i.e., to encode at least one command for each

desired prosthesis operation (grasping modes, wrist rotation, combined grasping modes, calibration. . .).

In MANUS-HAND, each individual finger behaves as a virtual spring with its two ends attached respectively to the palm and the object. In such a configuration, provided enough contact points are present, the system has stable equilibrium poses.

Several advantages can be reported. First, some adaptation to the object stiffness is autonomously provided. The same adaptation is present with regard to object environment interaction. Because the whole hand-object system has stable equilibrium poses, the computation of equilibrium configurations based on contact information is not needed.

On the other hand, stiffness controlled grasps require continuous motor operation with the subsequent energy consumption. In order to take advantage of the active compliance approach without increasing the energy consumption the following approach is studied:

- (a) First, all fingers' compliance is controlled according to the commanded grasping mode.
- (b) Secondly, once the equilibrium position is reached, all active fingers are blocked, thus maintaining the grasp without energy consumption. This approach would result in low robustness in the case of high object-hand interface stiffness. In our case, the interface stiffness is relatively low due to the force sensor low stiffness (see Section 4.2), the compliant crossed tendon transmission and the use of rubber gloves.

The following paragraphs are devoted to the Human Machine Interface, the control electronics and strategy.

4.1. Human Machine Interface

The command interface is based, within the MANUS-HAND approach, on the EMG residual signals of the amputee. Commercial EMG-controlled prostheses have a maximum of two independently—but not simultaneously—controlled active joints, see Otto Bock System Electric Hand (2001). Usually, flexors and extensors of the forearm are used to control every axis of these hands, a sudden simultaneous contraction of both flexors and extensors is used to shift the control from one axis to the other one. Due to the complexity of MANUS-HAND, this shifting scheme is not practical.

Pattern recognition has been attempted in the context of EMG signals for a variety of applications. EMG

patterns were used by Han et al. (2000), to assist elderly people in controlling a rehabilitation robotic arm. They implement pattern recognition techniques based on fuzzy logic and set theory tools to identify a set of EMG essential features with independence of the user. A scheme based on fuzzy pattern classification is then employed to handle the selected set of features. A similar scheme is reported in Chan et al. (2000). EMG management has been used in combination with tactile information to provide means for triggering state transition, see Kyberd and Chappel (1994).

We have selected a different approach also based on pattern recognition techniques. One single muscle EMG signal is used to encode the grasping commands. To this purpose, a command language was developed. Each word in the command language comprises three EMG bits (muscle contractions). In turn, each bit can adopt three digital levels “1”, “2” or “3” (corresponding to the amplitude of the EMG signals). As a consequence, a command comprises three muscle contractions in such a way that theoretically up to 27 different command could be available.

From a practical point of view, just 18 of these 27 commands can be implemented. This is because it is impossible to distinguish “0XY” from “XY0” patterns. In addition, in order to improve interference rejection the “X00” pattern was also eliminated from the command language, so filtering out spurious EMG interference peaks. Figure 7 shows a typical command word implemented by one of the amputees during clinical trials.

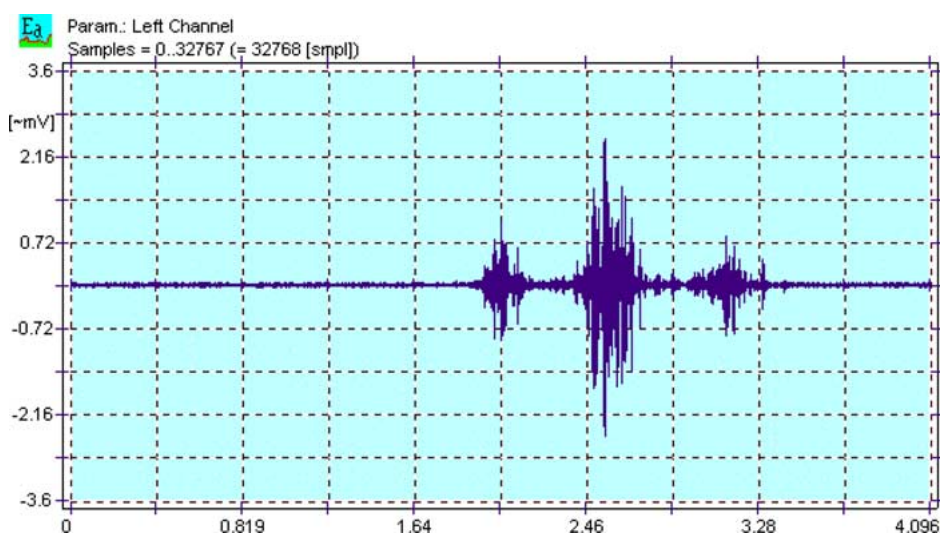


Figure 7. Typical EMG command word: EMG voltage vs. time.

4.2. Control Electronics

The control electronics gives support to the three main operations performed by the prosthesis:

1. EMG signal acquisition and command decoding
2. Autonomous control of grasps
3. Force biofeedback

EMG disposable pick-ups are located on the amputee muscle where enough residual activity is found. In addition force and position sensors are distributed in the hand. A position sensor is used to determine the angular position of each active axis. Likewise, force sensors are distributed at the finger tips (index and middle) and at the thumb tip. Both kinds of sensors were specifically designed from Hall-effect pick-ups.

Position sensors were developed by placing a permanent magnet opposite to the Hall effect sensor. In between, a cam made of ferromagnetic steel modifies the reluctance of the magnetic circuit resulting in a linear relation between cam rotation and output voltage, see Fig. 8a. The force sensors are embedded in the finger tips and are also based on Hall effect pick-ups. In this case, the permanent magnet is spring mounted and the resulting magnetic field at the Hall effect sensor location results in a linear relationship between force exerted and output voltage. The reader is referred to Pons et al. (2001) for a more detailed explanation of these sensor design.

A block diagram of the general hierarchical electronic architecture is depicted in Fig. 9. The electronic architecture comprises a host microcontroller (master) and three local controllers (slaves). They are linked through a fast I2C bus that allows bi-directional communication between master and slaves. Each local

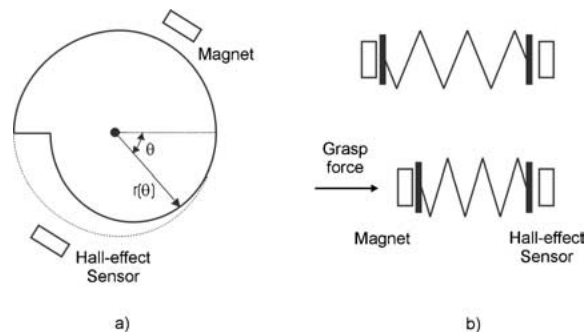


Figure 8. Schematic representation of (a) position and (b) force sensors based on Hall effect pick-ups.

controller is in charge of the active compliance control of each active joint. The control strategy is detailed in the next Section. The Host controller is also in charge of the EMG command decoding and the force biofeedback. The latter is introduced to partially overcome the inherent lack of feedback capabilities of EMG command approaches. A vibration device is attached for this purpose to the prosthesis socket. The vibration frequency is made proportional to the grasp force so that the user can have some feedback information on the force condition of the hand. When the system is initialised the Host controller performs the EMG calibration. In this procedure, particular EMG parameters (noise, thresholds, maximum time of bits . . .) are obtained.

After this calibration the Host controller is in sleep mode until the EMG signal passes a pre-established threshold. Once this is detected, the command is computed and, if valid, control set points are passed to the local controllers.

The local controllers are in charge of performing the low level programmable finger compliance control as explained in more detail in the next paragraph. This operation is performed according to the position and force information obtained from the sensors distributed in the hand. This information is also fed back to the Host controller to implement the force biofeedback to the user. The control architecture includes specific drivers for both the actuators and the sensors.

4.3. Control Strategy

In this section we introduce the compliance control of the active joints. As described in Section 3.1, for grasp control we have two active fingers (index and middle) and the thumb, see Fig. 10a. Both fingers implement the underactuated principle and just one actuator is used to drive the index and middle fingers. Fig. 10a shows schematically the active compliance approach. For every grasping mode, the finger performance is determined by a set of desired positions and finger stiffnesses (set points).

As a first approach, set points for each grasp are considered the full closed position for that particular grasp. For instance, if cylindrical grasps are considered, the set point would be full flexion for index and middle fingers and full flexion in opposition for the thumb.

The force exerted by each finger in this grasp obeys Hooke's law, thus behaving as a spring. Now suppose one of the fingers touches the object and starts applying

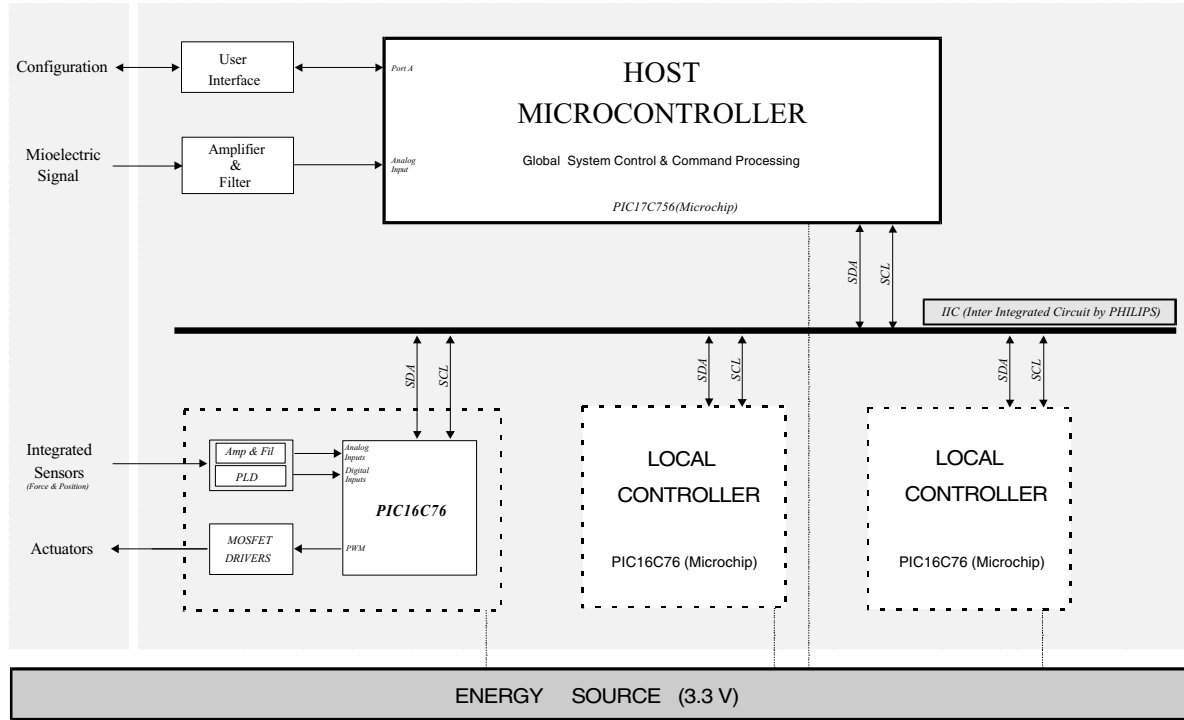


Figure 9. Hierarchical control electronic architecture, a master controller for EMG management and definition of grasp set points and three slave controllers to implement low level stiffness control of the hand.

force on it. If the force exerted on the object is too high, the finger will move the object against the opposite thumb but the exerted force will decrease because now the finger is closer to the desired position. Therefore, some adaptation to the object stiffness is present. Eventually equilibrium is reached in which finger and thumb forces match each other.

The control scheme is shown in Fig. 10b. Each local controller independently implements this control scheme with the grasp information provided by the host controller according to the user command. The host set points specify desired neutral position for the equivalent spring as well as spring rate. The spring rate controls the grasp force while the desired position mainly affects the grasp mode.

Table 5 presents the desired positions and spring rates to be commanded for every grasping mode. Values for these position set points are normalised so that 1 means full extension, 0 means full flexion and 1/2 means the thumb neutral position corresponding to the change of motion plane as explained in Section 3.1.

Figure 11 shows the position and force sensor readings during a typical cylindrical grasp. While a quite

Table 5. Control set points for local controllers as a function of grasp mode. The subscript “ind” refers to the index finger, “thf” refers to thumb and “wr” refers to wrist.

Position set point	Spring rate set point
Cylindrical grasp	
$\theta_{\text{ind}} = 0$	K_{dind}
$\theta_{\text{thf}} = 0$	$2 K$
$\theta_{\text{wr}} = ?$	
Precision grasp	
$\theta_{\text{ind}} = 1/2$	K_{dind}
$\theta_{\text{thf}} = 1/2$	K
$\theta_{\text{wr}} = ?$	
Hook grasp	
$\theta_{\text{ind}} = 0$	K_{dind}
$\theta_{\text{thf}} = 0$	K
$\theta_{\text{wr}} = ?$	
Lateral grasp	
$\theta_{\text{ind}} = 0$	$K = \infty$
$\theta_{\text{thf}} = 0$	K
$\theta_{\text{wr}} = ?$	

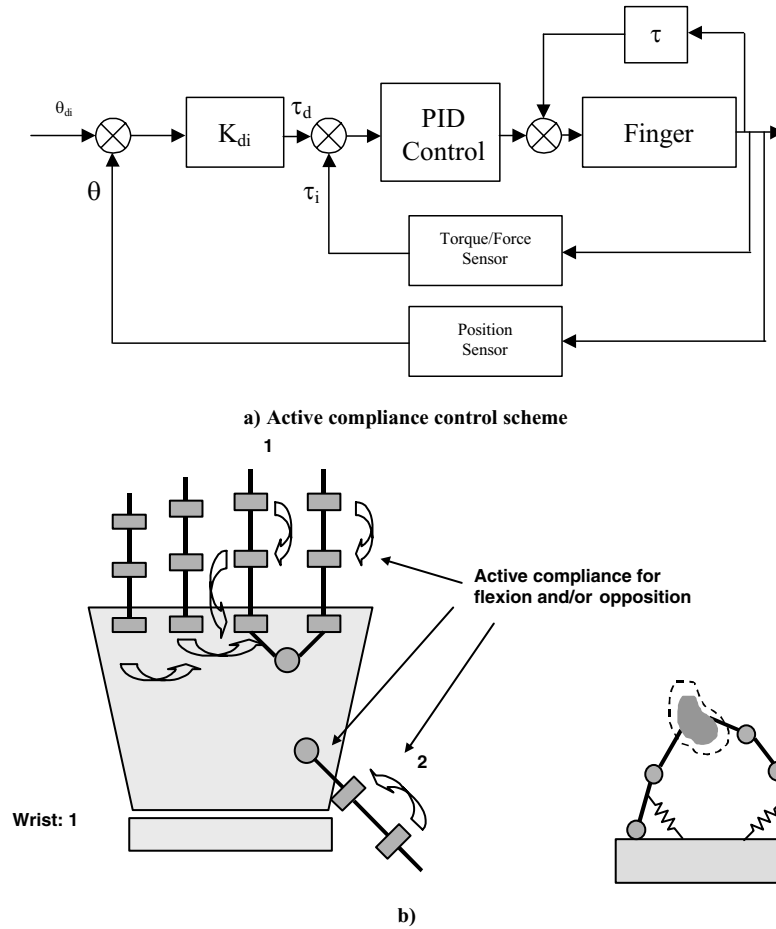


Figure 10. Active compliance approach implemented in MANUS-HAND: (a) basic stiffness control strategy and (b) schematic representation of the virtual spring concept.

smooth position response is obtained, it can be seen that a relatively high force overshoot just after contact was reached. However, it was experienced that force stabilization was quick enough to allow a time delayed blocking of actuators after contact is reached. In addition, no separation was observed between contacting surfaces.

When the system is started a calibration procedure is implemented. The master controller acquires EMG signals during 10 seconds at a sampling rate of 400 Hz to determine the noise level. The user is asked to repeat the command “121” to determine EMG thresholds.

After calibration EMG is acquired at 5 kHz and the master controller keeps recognizing valid commands. Whenever a valid command is detected, set points (position and spring rate) corresponding to that particular grasping mode are sent to slave controllers. Slave controllers implement the stiffness control loop at 500 Hz.

Force and position sensor signals are sampled at 10 kHz and averaged, so that an effective sampling frequency of 800 Hz is obtained. The resulting position sensor accuracy was close to 1 degree. As far as the force sensors are concerned, several spring rates were tried. The most common configuration allowed a force range of 60 N and a resolution of 1 N.

The slave controller analogue output is used to drive the ultrasonic motors. DC motors are driven by PWM output signals available at the slave controllers. The reference frequency of this PWM signals was set to 20 kHz.

5. Results and Discussion

The validation of the MANUS-HAND system was carried out on the prosthesis prototype and on a Virtual

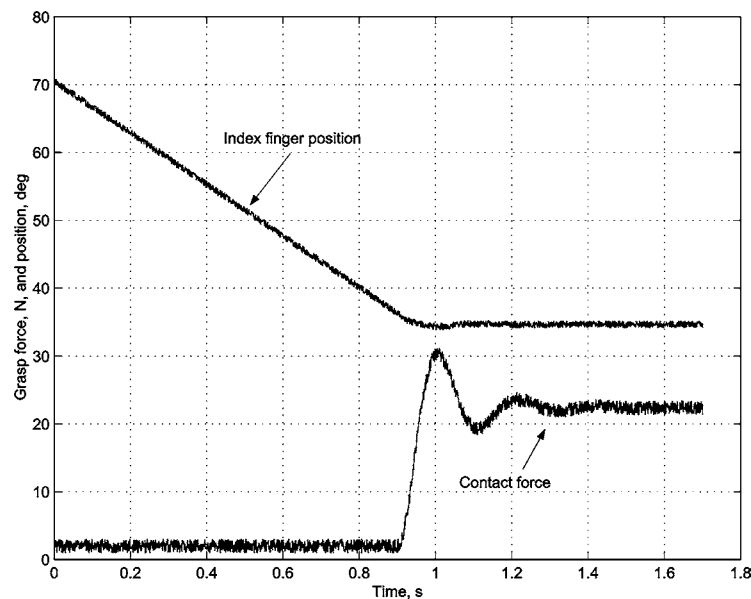


Figure 11. Force and position sensor readings during a typical power grasp. A sudden increase in force is observed after contact but no separation between contacting surfaces was experienced.

Reality, (VR), training platform. This validation process involved fifteen amputees. It was conducted in Spain, at the Centro de Recuperación de Minusválidos Físicos and in Israel, at the Flieman Hospital. All individuals gave informed consent to the experimental procedures.

The validation plan was structured in several stages, namely, first, a validation group selection; secondly, learning of EMG management, thirdly, learning of the digital EMG language, next, association of commands with grasping modes and active pronation-supination and finally, to generalise these grasping modes in basic activities of daily living.

The first four steps were developed both with the VR training platform and with the prosthesis prototype. The details of the validation of the command language are given in more detail in Pons et al. (2001). Just as a summary, we found that the three bit command language was easily learned by the users. After a few minutes of training the users were able to generate valid commands. It was observed that some patterns were more difficult than others. In particular, those patterns in which a high EMG amplitude bit is followed by a lower EMG amplitude bit, i.e., “211” resulted in added difficulty. This led us to associate easier commands to most common grasping modes.

The time required to perform a valid command varied greatly from user to user. Commanding times as

short as 0.4 s were observed for the most experienced users while some users only reached 2.5 s. All the amputees were able to generate at least 12 commands of the three-bit language without errors, and all were able to achieve the four grasping modes as well as the pronation and supination of the wrist. All the amputees were enthusiastic about the novelty of the prosthesis, with the extended range of functions and the similarity to a natural hand.

The generalization of these grasping modes in activities of daily living was done at the Spanish center, once the final integration of the Prosthesis was concluded. We had some days of co-ordinated work between the technical developers and the co-ordinators of the evaluation tests regarding the use of the whole-integrated system.

We used a provisional socket, adjustable with Velcro and with a supracondylar extension, getting a good fixation and adaptable to several stumps as was required for the validation.

The first subject of the evaluation was a man, 45 years old, with a congenital ectromelia and a long stump, see Fig. 12. He had never used a prosthesis before and he also collaborated in the VR training platform validation. He was able to generate fifteen commands of the “three-bit language” of MANUS-HAND without difficulties. There was a good fixation degree with the provisional socket, and after some training the amputee



Figure 12. The MANUS-HAND prototype during clinical trials. The prosthesis is fitted to one amputee who is using the hand for grasping a pen.

was able to get the precision, cylindrical, and lateral and hook grasping modes.

He liked MANUS-HAND appearance, and although at the beginning he found MANUS-HAND Prototype a little heavy, in a few minutes he was used to wear the prototype and felt really comfortable with it. This amputee collaborated in the validation during several days and the results were quite satisfactory. Two other amputees, males, 32 and 28 years old and with a trans-radial amputation of traumatic origin were engaged later on to continue validation of MANUS-HAND prototype with similar results.

The results were in line with the specifications. The resulting prototype was approximately 20% larger than an average male human hand. The resulting weight including socket was 1.2 kg, this is in fact over the specifications but there is margin for weight savings in future prototypes through reduction of size and a more optimised use of materials.

A maximum grasping force of 60 N was obtained at the finger tip. This is relatively lower than commercial hands (for instance, Otto Bock grippers provide up to 100 N) but the requirements on a hand with interphalanx active joints are less stringent than for fixed shape fingers (as found in all commercial upper limb prostheses). In order to take advantage of all the available motor power, the re-design of cosmetic gloves must be considered. Commercial cosmetic gloves are only appropriate to fixed shape fingers. To be used with adaptive fingers, more compliant glove materials must be used at the articulations.

Four basic grasping modes were successfully performed, i.e., precision, power, lateral and hook. In addition wrist rotation was also provided as specified. Spare control channels were successfully used to implement combined grasping modes. All grasps were fully reached in less than 1.2 s when implemented at the maximum available motor speed.

Energy consumption was one of the most relevant problems experienced with the MANUS-HAND prototype. An average operational time of four hours on a single battery charge was found. Since at the time the system was tested there was no approval for Li-ion batteries for medical applications, NiCd batteries were used. Currently Li-ion technology is a standard one in upper limb prosthetics, therefore based on the higher volumetric energy density of this technology some improvement might be expected. Nevertheless, an optimization on energy consumption is in order for future prototypes.

The hand operation was in general silent, in particular the wrist rotation. The thumb operation however was slightly noisy due to transmission backlash and play. It was concluded that a redesign of this particular mechanism was required to improve noise.

6. Conclusions

In this paper we have introduced an outline of the main details of the mechanical design and control aspects of the MANUS-HAND prototype. The paper shows the methodology implemented in our approach, i.e., from user specification to system design and manufacturing and clinical validation.

The proposed mechanical concept introduces a number of innovations to the field of currently existing upper limb prosthesis. The number of active joints is more than tripled as compared to commercial prostheses and up to four grasping modes plus wrist rotation are available. In our development, we adopted an underactuated kinematics concept to keep system complexity, size and weight within reasonable limits. This underactuated principle lead to cross-tendon coupling of fingers' phalanges and to the application of the Geneva mechanism to enable thumb flexion and opposition with just one single actuator.

The control approach takes into consideration the reduced bandwidth of Human Machine Interfaces, (HMI). Nevertheless, a new concept for the management of EMG signals is also introduced. This enabled an active compliance control of the fingers and the

thumb for autonomous control of grasp. In this scheme, just a couple of set points (position and spring rate) define the grasp, while low level control is performed by local controllers.

The clinical evaluation results show a good acceptance of the proposed system. The users of the MANUS prototype successfully implemented the four required grasping modes and a set of combined hand operations. Some minor changes are required to obtain increased sensitivity for some of the integrated sensors plus some adjustments to the hand kinematics.

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